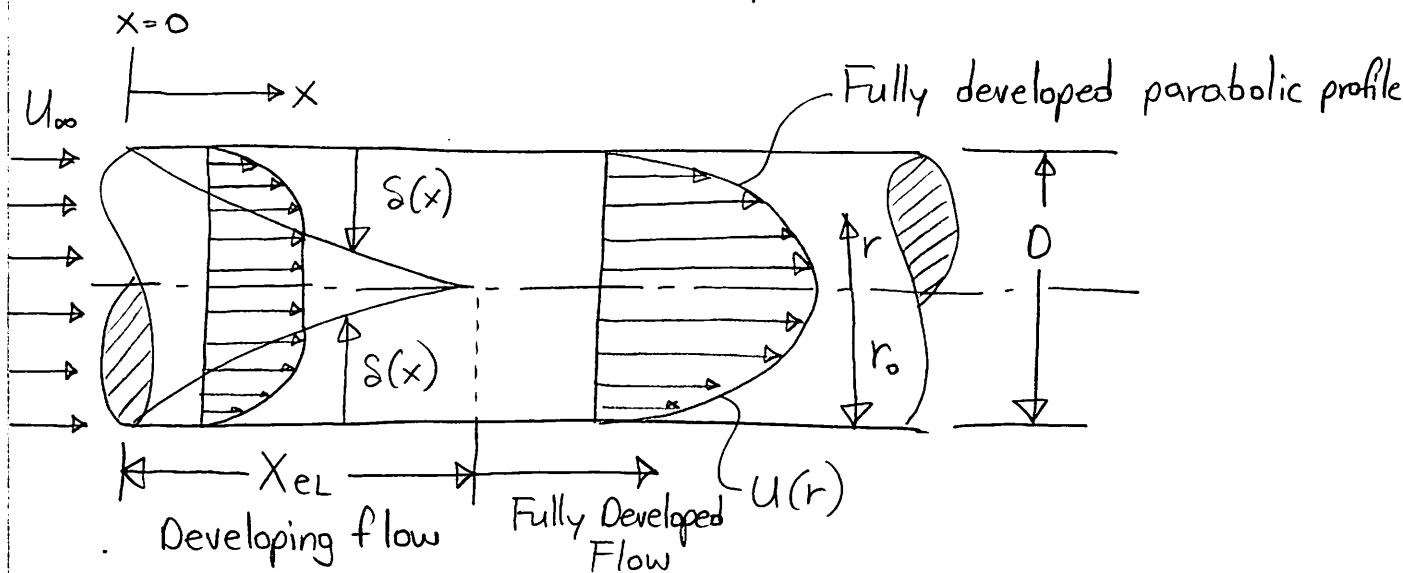


Internal Flow - Fully Developed Flow in Tubes



X_{eL} $\equiv$ entrance length or developing length. Here, the velocity profile varies with radial position r , and axial location, x .

We can estimate the X_{eL} by using our Blasius solution to see when our boundary layers begin to overlap:

$$\frac{\delta}{x} = \frac{5.0}{\sqrt{Re_x}}$$

We can estimate that $\delta = \frac{D}{2}$ when b.l.'s merge

scales with

$$\frac{D}{2X_{eL}} \sim \frac{5.0}{\sqrt{Re_x}}$$

$$\frac{D}{X_{eL}} \sim \frac{10}{\sqrt{Re_x}} = \frac{10}{\sqrt{\frac{\rho U_\infty X_{eL}}{\mu}}} = \frac{10}{\sqrt{\frac{\rho U_\infty X_{eL}}{\mu} \cdot \frac{D}{D}}} = \frac{10}{\sqrt{\frac{\rho U_\infty D}{\mu}} \cdot \sqrt{\frac{X_{eL}}{D}}}$$

$$\sqrt{\frac{D}{X_{eL}}} \sim \frac{10}{\sqrt{Re_D}} \Rightarrow \boxed{\frac{X_{eL}}{D} \sim \frac{Re_D}{100} \sim 0.01 Re_D} ; Re_D = \frac{\rho U_\infty D}{\mu}$$

Experimentally: $\boxed{\frac{X_{eL}}{D} = 0.05 Re_D}$: If $x > X_{eL}$, flow is fully developed. (124)

Fully Developed Region ($x > x_{eL}$)

We need to solve for the hydrodynamic properties of the flow to help us understand the heat transfer:

Applying the axial momentum equation (x-momentum)

$$\rho \left(\frac{\partial u_x}{\partial t} + u_r \frac{\partial u_x}{\partial r} + \frac{u_\phi}{r} \frac{\partial u_x}{\partial \phi} + u_x \frac{\partial u_x}{\partial x} \right) = - \frac{\partial p}{\partial x} \\ + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_x}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_x}{\partial \phi^2} + \frac{\partial^2 u_x}{\partial x^2} \right] + \rho g_x = 0$$

For fully developed flow: $\frac{\partial u_x}{\partial x} = \frac{\partial u_x}{\partial t} = \frac{\partial u_x}{\partial \phi} = 0$
 $u_\phi = u_r = 0$

For flow in a horizontal tube: $g_x = 0$

Most of our terms drop out and we are left with:

$$- \frac{\partial p}{\partial x} + \mu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_x}{\partial r} \right) = 0 \quad (1)$$

B.C.'s: $\left. \frac{\partial u_x}{\partial r} \right|_{r=0} = 0$ (Symmetry) (2)

$$u_x(r=r_0) = 0 \quad (\text{no slip}) \quad (3)$$

From here on in, I will use $u_x = u$ (dropping x subscript)

Integrating eq. (1) twice & applying (2) & (3):

$$\boxed{u(r) = \frac{r_0^2}{4\mu} \left(- \frac{\partial p}{\partial x} \right) \left(1 - \frac{r^2}{r_0^2} \right)} \Rightarrow \text{Fully developed velocity profile in a round pipe. (Laminar flow)}$$

Solving for our average velocity:

$$\bar{u} = \frac{1}{\pi r_o^2} \int_0^{r_o} u \cdot 2\pi r dr = \frac{r_o^2}{4\mu} \left(-\frac{2p}{2x} \right) \underbrace{\int_0^1 (1-\lambda) d\lambda}_{1/2}; \lambda = \frac{r^2}{r_o^2}$$

$$\bar{u} = \frac{r_o^2}{8\mu} \left(-\frac{2p}{2x} \right)$$

$$u(r) = 2\bar{u} \left(1 - \frac{r^2}{r_o^2} \right)$$

$\Rightarrow$ Fully developed laminar flow in a round pipe.

Typically, we need to know friction, pressure drop, or force.

$$f = \frac{\Delta P}{\left(\frac{L}{D}\right) \frac{1}{2} \rho \bar{u}^2} \equiv \text{Tube friction factor. Easy way to calculate pressure loss in tubes of length } L, \text{ diameter } D.$$

$$C_{f,x} = \frac{\tau_x}{\frac{1}{2} \rho \bar{u}^2} \equiv \text{Local tube friction coefficient. Related to the friction factor, } f.$$

Drawing a force balance diagram on a segment of our fluid in the pipe. $\sum F_x = 0$ since $u(r) = f(x) \equiv$ fully developed



$$\tau_o (2\pi r_o L) = \Delta P \pi r_o^2 = \frac{\Delta P D}{2}$$

$$4\tau_o = \frac{\Delta P}{\left(\frac{L}{D}\right)}$$

We can say: $\boxed{4C_f = f}$ $\Rightarrow$ Let's now solve for friction factor (126)

Laminar flow

$$f = \frac{\Delta P}{\left(\frac{L}{D}\right) \frac{1}{2} \rho \bar{U}^2} \quad (1)$$

But we've just solved for laminar flow that: $\bar{U} = \frac{r_0^2}{8\mu} \left(-\frac{2P}{2x}\right)$

$$\frac{8\mu \bar{U}}{r_0^2} = -\frac{2P}{2x} \approx \frac{\Delta P}{L} \quad (2) \quad \begin{array}{l} \text{substitute here} \\ \downarrow \end{array}$$

From (1): $\frac{\Delta P}{L} \cdot \frac{2D}{\rho \bar{U}^2} = f = \frac{8\mu \bar{U}}{r_0^2} \cdot \frac{2D}{\rho \bar{U}^2}$

$$\frac{16\mu D}{r_0^2 \rho \bar{U}} = f = \frac{16\mu D}{\left(\frac{D}{2}\right)\left(\frac{D}{2}\right) \rho \bar{U}} = \frac{64\mu}{\rho \bar{U} D} = \frac{64}{Re_D}$$

$$\boxed{f = \frac{64}{Re_D}} \Rightarrow \text{Pipe friction factor for laminar flow in a smooth round tube. Darcy-Weisbach equation.}$$

We can also say the following:

$$\boxed{f \cdot Re_D = \text{constant}}$$

For non-circular channels/pipes:

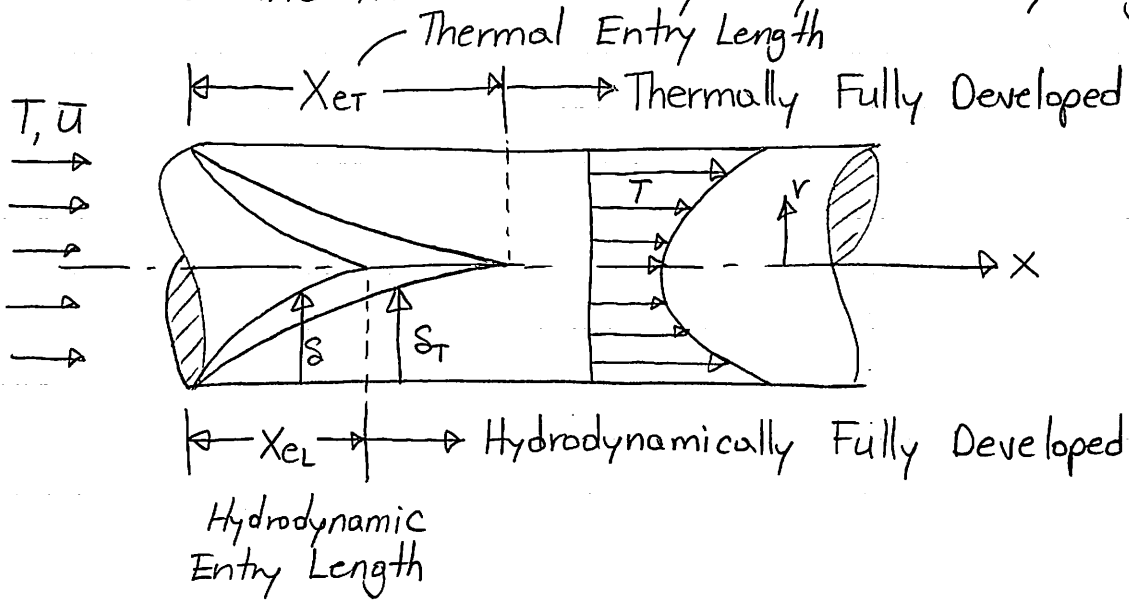
$$\boxed{D_h \equiv \text{hydraulic diameter} = \frac{4A}{P}}; \quad \begin{array}{l} A \equiv \text{cross sectional area} \\ P \equiv \text{wetted perimeter} \end{array}$$

Example

$$\left. \begin{array}{l} \text{Diagram of an equilateral triangle with side length } a. \\ A = \frac{a \cdot a \sin 60^\circ}{2} \\ P = 3a \end{array} \right\} D_h = \frac{4a^2 \sin 60^\circ}{3a} = 1.15a$$

Heat Transfer in Internal Flows

How does the thermal boundary layer or entry length develop?



For fully developed flow, both the Thermal entry length (X_{eT}) and hydrodynamic entry length (X_{eL}) must be passed: $X > X_{eT} \& X_{eL}$.

$$\boxed{\frac{X_{eT}}{D} = 0.017 Re_0 Pr} \Rightarrow \text{Experimentally obtained Thermal Entry Length}$$

So how about heat transfer in the fully developed region: i.e. $X > X_{eL} \& X > X_{eT}$

$$h = \frac{q''_{wall}}{\Delta T}$$

From energy balance at the wall (conduction = convection)

$$h \Delta T = k_f \frac{\Delta T}{\delta_T} \Rightarrow h \sim \frac{k_f}{\delta_T}$$

For fully developed flow (overlapping thermal boundary layers): $\delta_T \sim \frac{D}{2}$

$$h \sim \frac{2k_f}{D}$$

But we know that: $Nu_0 = \frac{hD}{k_f} = \frac{2k_f}{\cancel{D}} \left(\frac{\cancel{D}}{\cancel{k_f}} \right)$

$\boxed{Nu_0 \approx 2} \Rightarrow$ We expect a constant Nusselt #!

It turns out this scaling is pretty good. If we solve the coupled momentum and energy equations for internal smooth round tube pipe flow, we will obtain the following:

$$\boxed{\overline{Nu_0} = \frac{\bar{h}D}{k_f} = 4.364}$$

$\Rightarrow$ Constant wall heat flux
Laminar flow, round tube
 $Re_0 < 2300$; $Re_0 = \frac{\rho \bar{u} D}{\mu}$

$$\boxed{\overline{Nu_0} = \frac{\bar{h}D}{k_f} = 3.66}$$

$\Rightarrow$ Constant wall temperature
Laminar flow, round tube
 $Re_0 < 2300$

Bulk Fluid Temperature

It is important to note that for internal flows, heat transfer is computed by using the bulk fluid temperature: (T_b)

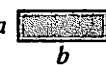
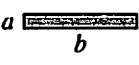
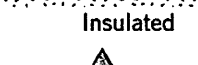
$$\bar{h} = \frac{q''|_{\text{wall}}}{T_w - T_b} \quad ; \quad \begin{array}{l} T_w = \text{wall temperature} \\ T_b = \text{bulk fluid temperature} \end{array}$$

T_b can be thought of as if we allow the pipe fluid to uniformly mix & come to an equilibrium temperature:

$$\boxed{T_b = \frac{1}{A \bar{u}} \int_A u(r) \cdot T dA}$$

where $A \equiv$ cross sectional area
 $\bar{u} \equiv$ average fluid velocity

TABLE 8.1 Nusselt numbers and friction factors for fully developed laminar flow in tubes of differing cross section

Cross Section	$\frac{b}{a}$	$Nu_D \equiv \frac{hD_h}{k}$		Re_{D_h}
		(Uniform q''_s)	(Uniform T_s)	
	—	4.36	3.66	64
	1.0	3.61	2.98	57
	1.43	3.73	3.08	59
	2.0	4.12	3.39	62
	3.0	4.79	3.96	69
	4.0	5.33	4.44	73
	8.0	6.49	5.60	82
	∞	8.23	7.54	96
 Heated	∞	5.39	4.86	96
 Insulated				
	—	3.11	2.49	53

Used with permission from W. M. Kays and M. E. Crawford, *Convection Heat and Mass Transfer*, 3rd ed. McGraw-Hill, New York, 1993.